

Advanced Algorithms and Data Structures

Hashing

Exercises

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1. Suppose you use open hashing (chaining) and the following keys are inserted: 5, 28, 19, 15, 20, 33, 12, 17, 10 and $m = 9$. In which slots do collisions occur?
2. Now suppose you use closed hashing with linear probing and the same keys as above are inserted. More collisions occur than in the previous question. Where do the collisions occur and where do the keys end up?
3. With quadratic probing and $m = 16$, where does the first collision occur?
4. Suppose n keys are chosen randomly and the hash function distributes the keys uniformly at random over $\{0, 1, \dots, m-1\}$.
 - a. What is the probability that $n=2$ independently chosen keys have the same hash value i.e. what is the probability there is a collision among those two keys?
 - b. What is the probability that there are no collisions among any of the n keys? Assume $n \leq m$.
5. Demonstrate what happens when we insert the keys 624, 373, 5, 409, 193, 184, 931, 17, 31, 100 into a hash table with collisions resolved by chaining. Let the table have 10 slots, and let the hash function be $h(k) = k \bmod 10$.
6. Consider inserting the keys 10, 22, 31, 3, 15, 28, 17, 88, 59 into a hash table of length $m=11$ using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using linear probing, and using double hashing with $h_1(k)=k$ and $h_2(k) = 1 + (k \bmod (m-1))$.
7. Given the input $\{42, 39, 57, 3, 18, 5, 67, 13, 70, 26\}$, and a fixed table size of 13, and a hash-function $H(X) = X \bmod 13$, show the resulting by using linear probing and separate chaining hash-table.
 - a. Linear probing hash-table
 - b. Separate chaining hash-table
8. Given the input $\{4371, 1323, 6173, 4199, 4344, 9679, 1989\}$, and a fixed table size of 10, and a hash-function $H(X) = X \bmod 10$, show the resulting
 - a. Separate chaining hash-table
 - b. Linear probing hash-table
 - c. Quadratic probing hash-table

9. Given the input {25, 13, 37, 42, 47, 22, 57, 3, 14, 1}, and a fixed table size of 11, and a hash-function $H(X) = X \bmod 11$, show the resulting
- Separate chaining hash-table
 - Linear probing hash-table
 - Quadratic probing hash-table
 - Double hashing where $H_2(X) = 1 + (X \bmod 9)$